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Executive Summary

- 5+ years of professional experience in information technology as Data Engineer with an expert hand in the areas of Database Development, ETL Development, Data modelling, Report Development and Big Data Technologies.
- Experience in Data Integration and Data Warehousing using various ETL tools Informatica PowerCenter, AWS Glue, SQL Server Integration Services (SSIS).
- Experience in Designing Business Intelligence Solutions with Microsoft SQL Server and using MS SQL Server Integration Services (SSIS), MS SQL Server Reporting Services (SSRS) and SQL Server Analysis Services (SSAS).
- Experience working with Amazon Web Services (AWS) cloud and its services like Snowflake, EC2, S3, RDS, EMR, VPC, IAM, Elastic Load Balancing, Lambda, RedShift, Elastic Cache, Auto Scaling, Cloud Front, Cloud Watch, Data Pipeline, DMS, Aurora, ETL and other AWS Services.
- Extensive experience in Data Mining solutions to various business problems and generating data visualizations using Tableau, PowerBI, Alteryx.
- Well knowledge and experience in Cloudera ecosystem such as HDFS, Hive, SQOOP, HBASE, Kafka, Data pipeline, Data analysis and processing with Hive SQL, IMPALA, SPARK, SPARK SQL.
- Experienced in design, development, Unit testing, integration, debugging and implementation and production support, client interaction and understanding business application, business data flow and data relations.
- Using Flume, Kafka and Spark streaming to ingest real time or near real time data to HDFS.
- Analysed data and provided insights with Python Pandas.
- Worked on AWS Data Pipeline to configure data loads from S3 into Redshift.
- Worked on Data Migration from Teradata to AWS Snowflake Environment using Python and BI tools like Alteryx.
- Developed Python scripts to parse the Flat Files, CSV, XML, JSON files and extract the data from various sources and load the data into data warehouse.
- Developed Automated scripts to do the migration using Unix shell scripting, Python, Oracle/TD SQL, TD Macros and Procedures.
- Excellent interpersonal and communication skills, experienced in working with senior level managers, business people and developers across multiple disciplines.
- Strong problem solving, analytical and have the ability to work both independently and as a team. Highly enthusiastic, self-motivated and rapidly assimilate with new concepts and technologies

Technical Skills

Programming & Scripting	Python, SQL, Java, Scala, PL/SQL, Shell Scripting, Terraform, Bash
Big Data & Distributed Computing	Hadoop, Spark, PySpark, Hive, HBase, Kafka, Sqoop, Flume, Presto, Delta Lake
Cloud Platforms & Services	AWS (S3, Redshift, EMR, Lambda, RDS, EC2, Glue, Athena), GCP (Big Query, Dataflow, Pub/Sub), Azure Data Factory
ETL & Data Pipelines	Apache Airflow, AWS Glue, Informatica, Talend, SSIS, DBT, Snowpipe
Databases & Warehousing	Snowflake, PostgreSQL, MySQL, SQL Server, Oracle, Teradata, MongoDB, DynamoDB
DevOps & Infrastructure	Terraform, Kubernetes, Docker, CI/CD (Jenkins, GitHub Actions)
Data Visualization & Analytics	Tableau, Power BI, Matplotlib, Seaborn

Work Experience

Client: PNC, OH **Nov 2024 – Present**
Role: Data Engineer

Responsibilities:

- Experience in building and architecting multiple Data pipelines, end to end ETL and ELT process for Data ingestion and transformation.
- Worked extensively with AWS services like EC2, S3, VPC, ELB, Auto Scaling Groups, Route 53, IAM, CloudTrail, CloudWatch, CloudFormation, CloudFront, SNS, and RDS.
- Design and Develop ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, Parquet/Text Files into AWS Redshift.
- Build a program with Python execute it in cloud Dataflow to run Data validation between raw source file and Big query tables.
- Strong background in Data Warehousing, Business Intelligence and ETL process (Informatica, AWS Glue) and expertise on working on Large data sets and analysis
- Building a Scala and spark based configurable framework to connect common Data sources like MYSQL, SQL Server.
- Develop and implement data engineering solutions using Python programming language.
- Utilize data manipulation libraries such as Pandas and NumPy for efficient data manipulation and analysis.
- Work with various database systems, including SQL and NoSQL databases like MySQL, PostgreSQL, MongoDB, etc.
- Used AWS data pipeline for Data Extraction, Transformation and Loading from homogeneous or heterogeneous data sources and built various graphs for business decision-making using Python matplot library
- Analyzed SQL scripts and designed the solutions to implement using PySpark.
- Worked on documentation of all worked Extract. Transform and Load, Designed, developed and validated and deployed the Talend ETL processes for the Data Warehouse team using Hive.
- Applied required transformation using AWS Glue and loaded data back to Redshift and S3.
- Create meaningful visualizations using tools like Matplotlib, Seaborn, or Tableau to effectively communicate insights.
- Experience with data lakes like AWS S3 for storing vast amounts of raw data in its native format.
- Extensively worked on making REST API calls to get the data as JSON response and parse it.

Environment: Pandas, AWS Redshift, Snowflake, S3, Postgres, MS SQL Server, BigQuery, Python, Postman, Tableau, Unix Shell Scripting, EMR, GitHub

Client: Velocious Solutions Pvt Ltd, India **Dec 2019 – Nov 2021**
Role: Data Engineer

Responsibilities:

- Managed data storage and lifecycle policies on Amazon S3, reducing data retrieval costs and improving access times.
- Administered relational and data warehousing solutions with Amazon RDS and Amazon Redshift, optimizing performance and scalability.
- Utilized Amazon EMR to process large datasets using Hadoop frameworks, significantly reducing processing time.
- Implemented NoSQL database solutions with Amazon DynamoDB, handling high-throughput data and scalable storage needs.
- Developed RESTful APIs using Flask and Python, enabling seamless application integration and improving response times.
- Built and deployed scalable web applications using Flask, incorporating user authentication, database management, and third-party API integrations.
- Designed and managed ETL pipelines with AWS Glue, processing data from multiple sources and improving data integration efficiency.
- Executed ad-hoc queries on S3 data using Amazon Athena, facilitating on-demand data analysis and reporting.

- Integrated PySpark with AWS S3 for large-scale data ingestion and processing in distributed environments.
- Used Amazon SNS and SQS for robust notification and message queuing services, enhancing inter-service communication.
- Developed and maintained complex ETL pipelines, ensuring data transformation and loading processes met organizational requirements.
- Developed internal APIs to expose business logic for customer-facing applications and partner integration
- Applied data warehousing concepts on Amazon Redshift, optimizing storage with partitioning and indexing, and reducing query times.
- Set up and managed data lakes on AWS, ensuring efficient organization and accessibility of large-scale data repositories.
- Designed and optimized data models for both relational and NoSQL databases, enhancing data integrity and query performance.
- Integrated data from multiple sources, enabling comprehensive analysis and improving report generation efficiency.
- Utilized Apache Hadoop and Apache Spark for big data processing, handling extensive data transformations and computations.

Environment: Amazon S3, AWS Lambda, Amazon RDS, Amazon Redshift, Amazon EMR, Amazon DynamoDB, AWS Glue, Amazon Kinesis, Amazon Athena, Amazon SNS, SQS, Data Warehousing Concepts, Data Lakes (AWS), Apache Hadoop, Apache Spark, Hive, Presto, Databricks, Python, AWS CloudFormation

Client: Value Momentum Software Services Pvt. Ltd, India June 2017 – Nov 2019
Role: Software Engineer

Responsibilities:

- Managed the setup and configuration of Hadoop infrastructure, ensuring ongoing cluster maintenance and log file management for smooth operations.
- Created shell scripts to monitor Hadoop daemon services, quickly addressing issues to maintain uninterrupted service.
- Deployed Cloudera Manager for comprehensive monitoring, alerting, and implementing security measures in the Hadoop cluster.
- Utilized CloudWatch logs to monitor application logs and set up alarms to proactively track system health and trigger actions on exceptions.
- Facilitated data migration from Oracle and SQL Server to Hadoop using Sqoop, processing a variety of flat file formats.
- Designed and implemented Flume and Sqoop data pipelines to ingest customer behavioral data histories into HDFS for analysis.
- Built and optimized Hive tables, executed MapReduce jobs, and created Hive join queries for complex data analysis.
- Integrated Kafka and Spark streaming to process real-time data, enabling specific use cases for dynamic analytics.
- Configured Spark streaming to ingest real-time data from Kafka and store it in HDFS, enhancing the system's real-time capabilities.
- Developed and optimized Spark and Hive jobs to summarize and transform large-scale data for reporting and analytics.
- Packaged and deployed scalable PySpark applications using Docker, ensuring portability and environment consistency.
- Utilized Spark and Python for efficient testing and transformation of data, leveraging Spark DataFrames to convert unstructured data into structured formats.
- Automated end-to-end data pipelines using Oozie, ensuring timely execution of PySpark jobs for manufacturing data analysis.

- Integrated Flume for collecting and analyzing web log data, implementing automated scheduling to ensure timely execution of batch jobs and data transformations.

Environment: Hadoop, HDFS, YARN, MapReduce, Sqoop, Flume, Zookeeper, Ambari, Oozie, PIG, Hive, HBase, Spark Core, Spark SQL, Python, Kafka, Shell-Scripting, Git, Jira, Jenkins